

COTIVITI INTERN ASSESSMENT TOPIC 3 · CONTENT MANAGEMENT IN HEALTH CARE

From Written Policy to Self-Verified Rules

A self-critiquing LLM compiler for healthcare coverage policies

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Healthcare runs on prose and the prose keeps moving

Every paid or denied claim traces back to written policy

Federal NCDs and LCDs, commercial medical policies, coding manuals, and payer-provider contracts, all authored as prose.

The corpus drifts continuously, not on a calendar

NCD reconsiderations are ad hoc and take 6–9 months; CPT/ICD code sets refresh annually; commercial policies layer on top.

Every change is re-encoded by hand

Clinical SMEs and rules analysts re-read the document and rewrite the logic; slow, expensive, and easy to get subtly wrong.

\$23–31B

Estimated annual U.S. cost of prior-authorization activity

Health Affairs (Casalino et al., 2009)

88%

Of prior-auth transactions still partially or fully manual

CAQH Index, 2018

~30%

Of U.S. healthcare spending tied to administration

Health care in the United States (overview literature)

Three curves converged in the same 24 months

REGULATION

CMS-0057-F (Jan 2024)

Mandates FHIR prior-authorization APIs for regulated payers. HL7 Da Vinci CRD / DTR / PAS supply the implementation guides, but every one of them presupposes the coverage policy already exists in machine-readable form.

AI METHODS

Self-critique loops

Self-Refine (Madaan et al., 2023) and Reflexion (Shinn et al., 2023) show an LLM acting as generator, critic, and reviser delivers large accuracy gains with no additional training. The loop the POC hand-rolls this.

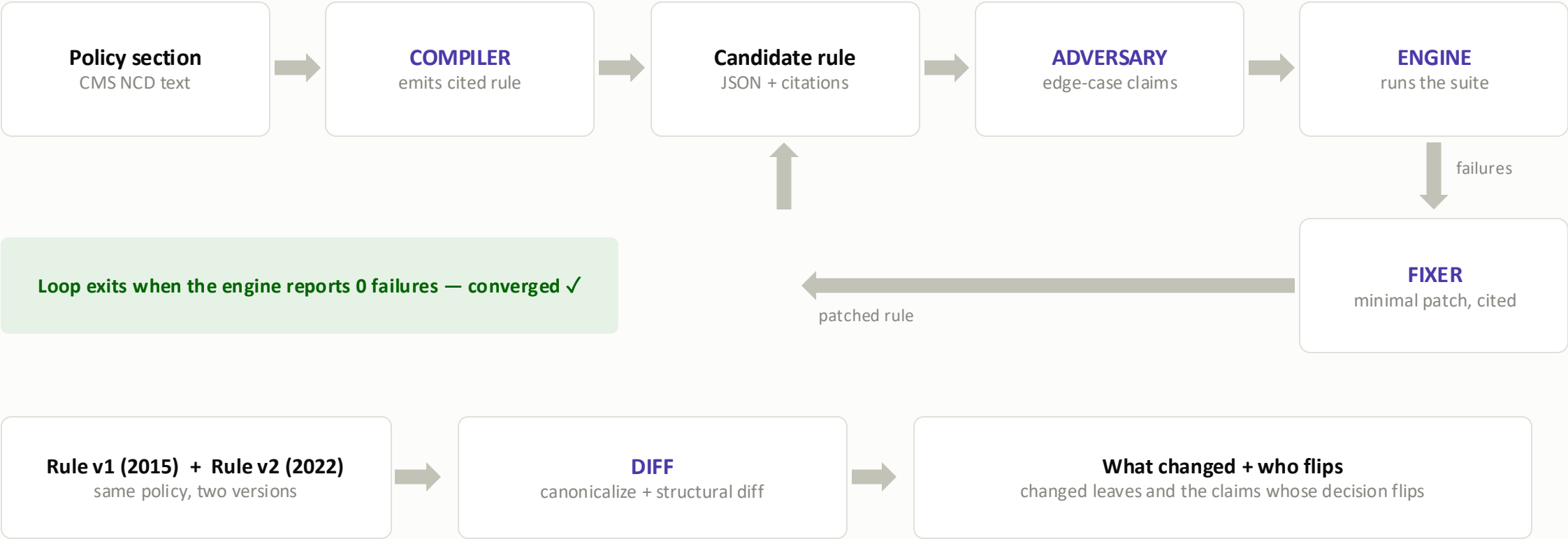
EXECUTION TARGET

Sandboxed rule formats

JSON Logic rules are data, not code: “no loops, no functions, no write access to anything.” A safe, portable compilation target an LLM can emit reliably, with CQL + Da Vinci DTR as the production path.

The standards are ready. Encoding the rules is the bottleneck.

policy-as-rule: compile, self-verify, diff



Guardrails: every leaf must cite a verbatim span of the source · policy silence routes to manual review, never a guess · loop capped at 5 rounds

NCD 210.14: lung cancer screening with LDCT, 2015 vs 2022

1 Compile the 2022 policy

The rule streams out as typed JSON. Every clause cites the exact sentence of the NCD it came from.

2 Seed a defect, watch the loop catch it

A deliberately wrong age threshold goes in; the Adversary's boundary claims expose it and the Fixer patches it — converged in 2 rounds.

3 Diff the 2015 and 2022 versions

Exactly three changed leaves: age 55→50, pack-years 30→20, LDCT code G0297→71271. Zero cosmetic noise.

4 The claim flip

One synthetic patient, who is 52 years old, with 22 pack-years, is evaluated live against both versions: DENIED in 2015, APPROVED in 2022.

→ switching to the running application (all claims are synthetic — no PHI)

What the demo proved

- ✓ **Self-verification works**
A seeded wrong age threshold was caught by the Adversary's boundary claims and patched by the Fixer. The loop converged in 2 rounds.
- ✓ **The diff is behavioral, not cosmetic**
Three modified leaves between 2015 and 2022: age, pack-years, LDCT code. Reviewers see what changed and who flips.
- ✓ **Citations ground every clause**
A rule with an uncited leaf is rejected before it ever runs; policy silence routes to manual review instead of a guess.

THE HEADLINE PATIENT · SYNTHETIC

52-year-old former smoker

22 pack-years · quit 8 years ago · asymptomatic

2015 policy (NCD 210.14 v1)

DENIED



2022 policy (NCD 210.14 v2)

APPROVED

Found automatically by running both rules over the adversarial suites

Three moves, in order of investment

1 EMBED

A compile-and-diff layer inside Payment Policy Management

As productivity multiplier for the existing analyst workforce, the SME becomes a reviewer, not a transcriber. Measure SME hours saved per policy onboarded and per revision.

2 EXPOSE

The diff artifact as a policy change feed on Edifecs' FHIR stack

A CMS-0057-F-aligned API that providers and downstream payers subscribe to, converting a back-office capability into a sellable interoperability product.

3 PUBLISH

A public benchmark on CMS NCDs and LCDs

Policy-pairs with SME-graded expected rule deltas. Plants the methodological flag and answers regulatory scrutiny with evidence rather than assurance.

Production path: JSON Logic (POC target) → CQL + Da Vinci DTR/CRD (the FHIR-native target Edifecs already speaks)

**Ship a self-verifying compiler with a diff feed,
not just another rules editor.**

Thank you

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Report, POC code, and this deck: [GitHub repository](#) (link in email)